

ApexPlanet Software Pvt. Ltd.

Data Analytics Internship Capstone

End-to-End Case Study Report

Data Wrangling • SQL & Python EDA • Power BI RFM Dashboard • Hypothesis Testing

Prepared by: Brojo Mohan Dutta

Konnagar, Hooghly, West Bengal, India

Report Date: September 2026

Analysis Period: January 2025 – January 2026 | Dataset: 1,000 Transactions

Table of Contents

Table of Contents.....	2
Executive Summary	4
Key Metrics at a Glance	4
1. Project Overview	5
1.1 Objective.....	5
1.2 Deliverable Structure.....	5
1.3 Technical Stack	5
2. Task 1 — Data Immersion & Wrangling.....	6
2.1 Data Dictionary	6
2.2 Data Quality Findings.....	6
2.3 Profiling Code	6
2.4 Cleaning & Transformation Pipeline.....	7
2.5 Output.....	7
3. Task 2 — Exploratory Data Analysis & Business Intelligence	9
3.1 Business Questions	9
3.2 SQL Findings.....	9
Monthly Revenue Trend	9
Top Products & Categories	9
City Performance	9
Customer Demographics	10
3.3 Core SQL Query — Monthly Revenue Trend	10
3.4 Core SQL Query — Month-over-Month Category Growth.....	10
3.5 Python Multivariate EDA (995 transactions, 2025 only)	11
Correlation Matrix	11
Pair Plot by Category	12
Scatter Plot — Age vs. Total Sales by Category	12
Scatter Plot — Quantity vs. Unit Price by Gender	13
Heatmap — Avg Order Value by City × Category	13
Heatmap — Avg Order Value by Age Group × Category	14
Box Plot — Total Sales by Category and Gender	14
Static Executive Dashboard Mock-Up.....	15
3.6 Section Summary	15
4. Task 3 — Deep-Dive Analysis & Interactive Dashboarding (Power BI)	16
4.1 RFM Methodology	16
4.2 Core DAX Measures	16
4.3 RFM Segment Profiles.....	16

Segment Strategic Profiles.....	17
4.4 Dashboard — Page 1: Executive Overview.....	17
4.5 Dashboard — Page 2: Customer Segmentation Deep-Dive	18
4.6 Slicers & Interactivity.....	18
4.7 Data Model Reconciliation	18
4.8 Strategic Recommendations Summary	19
5. Task 4 — Hypothesis Testing & Statistical Inference.....	20
5.1 Hypothesis 1 — Gender-Based Spending Difference	20
5.2 Hypothesis 2 — Age Group × Category Independence	20
Category Preference by Age Group (Normalized %)	21
5.3 Consolidated Statistical Verdicts	22
6. Conclusion & Strategic Recommendations.....	23
6.1 Recommended Next Steps	23
6.2 Portfolio Artifacts.....	23

Executive Summary

This report documents the complete, end-to-end Data Analytics Internship capstone delivered for ApexPlanet Software Pvt. Ltd., covering data wrangling, exploratory data analysis (SQL and Python), interactive Power BI dashboarding with RFM customer segmentation, and formal statistical hypothesis testing. The project transformed a raw 1,000-record transactional Sales_Dataset into a fully reconciled, decision-ready analytics product across four sequential deliverables (Task 1–Task 4).

The underlying business generated ₹139.4M in total revenue across 1,000 orders from 947 unique customers over the analysis window (January 2025 – January 2026), at an average order value of ₹139K. Electronics is the dominant category, contributing ₹50.78M (36.4% of revenue), followed by Education, Grocery, Furniture, and Fashion in a diversified long tail.

RFM (Recency, Frequency, Monetary) segmentation of the customer base surfaced material revenue-at-risk: Potential Loyalists (244 customers, ₹40.6M) are the largest single contributor but are trending toward dormancy, while At-Risk customers (238 customers, ₹35.4M) represent an active churn exposure requiring immediate win-back intervention. Champions (35 customers, ₹12.3M) form a high-value but highly concentrated core.

Formal hypothesis testing (independent t-test and Chi-Square test of independence) found no statistically significant relationship between gender and spending ($p = 0.495$) or between age group and category preference ($p = 0.837$) — both traditional demographic levers were rejected as segmentation strategies in favor of the empirically validated RFM lifecycle model.

Key Metrics at a Glance

Metric	Value
Total Revenue	₹139,399,440
Total Orders	1,000
Unique Customers	947
Average Order Value	₹139,399
Top Category	Electronics (36.4% of revenue)
Top City	Patna (₹19.29M)
Average RFM Score	7.50 / 12
At-Risk Revenue Exposure	₹35.4M (238 customers)
Champions Revenue Concentration	8.8% of revenue from 3.7% of customers

1. Project Overview

1.1 Objective

Build a complete data-analytics pipeline for ApexPlanet Software Pvt. Ltd.'s Sales_Dataset — from raw-file data quality assessment through to an interactive, stakeholder-ready Power BI decision-support tool — while validating every descriptive finding with formal statistical inference.

1.2 Deliverable Structure

- Task 1 — Data Immersion & Wrangling: profiling, cleaning, and feature engineering of the raw Excel source in Python/Pandas.
- Task 2 — Exploratory Data Analysis & Business Intelligence: BigQuery SQL analysis plus Python multivariate EDA and a static dashboard mock-up.
- Task 3 — Deep-Dive Analysis & Interactive Dashboarding: customer-level RFM segmentation and a two-page interactive Power BI dashboard.
- Task 4 — Hypothesis Testing & Statistical Inference: formal significance testing of demographic spending patterns.

1.3 Technical Stack

Component	Technology	Role
Raw Data	Excel (.xlsx) → CSV	ApexPlanet_DataAnalytics_Dataset
Transformation	Python / Pandas, NumPy	Data cleaning, derived fields
EDA & BI	BigQuery Standard SQL	Business-question queries, CTEs, window functions
Statistical Analysis	Python — SciPy, Seaborn, Matplotlib	Correlation, hypothesis testing, visualization
RFM Pipeline	SQL / BigQuery	Customer-level aggregation & NTILE(4) scoring
Dashboard	Power BI Desktop / Service	2 pages, 8 visuals, 6 slicers, 17 DAX measures

2. Task 1 — Data Immersion & Wrangling

This stage profiled and cleaned the raw ApexPlanet Sales_Dataset (1,000 transaction records, 12 source columns), sourced from ApexPlanet_DataAnalytics_Dataset.xlsx, sheet Sales_Dataset, executed in Jupyter Notebook 7.4.5.

2.1 Data Dictionary

Column	Type	Description
Order_ID	String	Unique transaction identifier (primary key)
Order_Date	Date	Order date, YYYY-MM-DD
Customer_ID	String	Unique customer identifier
Customer_Name	String	Customer name (PII, excluded from aggregate analysis)
Age	Integer	Customer age at time of purchase
Gender	String	Male / Female
City	String	Order city
Product	String	Product purchased
Category	String	Product category
Quantity	Integer	Units purchased
Unit_Price	Float	Price per unit (INR)
Total_Sales	Float	Quantity × Unit_Price

2.2 Data Quality Findings

Issue	Finding	Rows Affected
Missing values	Age missing	20 (2.0%)
Missing values	City missing	13 (1.3%)
Duplicates	Full-row duplicates	0
Duplicates	Order_ID collision (ORD100050)	9
Date formatting	Order_Date stored as text	1,000 (100%)
Outliers (IQR)	Across Age, Quantity, Unit_Price, Total_Sales	0

Key observation: Nine rows shared Order_ID “ORD100050” despite representing distinct customers, products, and dates — a broken upstream primary key, not a duplicate transaction. All values fell within plausible business ranges (Age 18–65, Quantity 1–10, Unit_Price ₹145.78–₹49,997.53), so no IQR outliers were removed.

2.3 Profiling Code

```
import pandas as pd
import numpy as np

RAW_PATH = "ApexPlanet_DataAnalytics_Dataset.xlsx"
df = pd.read_excel(RAW_PATH, sheet_name="Sales_Dataset")
```

```

print("Shape:", df.shape)
df.info()

# Missing values
missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)
print(pd.DataFrame({"Missing_Count": missing, "Missing_%": missing_pct})
      .query("Missing_Count > 0"))

# Duplicate records
print("Full-row duplicates:", df.duplicated().sum())
print("Rows sharing a non-unique Order_ID:",
      df["Order_ID"].duplicated(keep=False).sum())

# Outlier scan (IQR method)
def iqr_outliers(series):
    q1, q3 = series.quantile([0.25, 0.75])
    iqr = q3 - q1
    lower, upper = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    return ((series < lower) | (series > upper)).sum()

```

2.4 Cleaning & Transformation Pipeline

- Age (numeric): median imputation — robust to skew, avoids fabricating a value influenced by extremes.
- City (categorical): filled with explicit “Unknown” label rather than the modal city, to avoid inventing geographic signal.
- Full-row duplicates: none found; drop_duplicates() retained defensively for future loads.
- Order_ID collisions: NOT deleted. Each of the 9 rows was reassigned a unique surrogate suffix (-01, -02, ...) and flagged via a new Order_ID_Flag column, with the original preserved in Order_ID_Original — auditable rather than destructive.
- Order_Date: converted with pd.to_datetime(..., format="%Y-%m-%d", errors="coerce") to avoid ambiguous parsing.
- Feature engineering: Order_Year, Order_Month, Order_Month_Name, Age_Group (5 bands: 18-25 ... 56-65), and Avg_Price_Check (Total_Sales ÷ Quantity) as a QA reconciliation field.

```

# 3. HANDLE DUPLICATES
dupe_id_mask = df["Order_ID"].duplicated(keep=False)
df = df.reset_index(drop=True)
df["Order_ID_Original"] = df["Order_ID"]
df["Order_ID_Flag"] = np.where(dupe_id_mask, "Duplicate_ID_Reassigned", "OK")
df.loc[dupe_id_mask, "Order_ID"] = [
    f"{oid}-{i+1:02d}" for i, oid in enumerate(df.loc[dupe_id_mask, "Order_ID"])
]

# 4. STANDARDIZE Order_Date
df["Order_Date"] = pd.to_datetime(df["Order_Date"], format="%Y-%m-%d", errors="coerce")

# 5. FEATURE ENGINEERING
df["Order_Year"] = df["Order_Date"].dt.year
df["Order_Month"] = df["Order_Date"].dt.month
df["Order_Month_Name"] = df["Order_Date"].dt.month_name()
age_bins = [17, 25, 35, 45, 55, 65]
age_labels = ["18-25", "26-35", "36-45", "46-55", "56-65"]
df["Age_Group"] = pd.cut(df["Age"], bins=age_bins, labels=age_labels, include_lowest=True)
df["Avg_Price_Check"] = (df["Total_Sales"] / df["Quantity"]).round(2)

# 6. EXPORT
df.to_csv("cleaned_sales_dataset.csv", index=False)

```

2.5 Output

cleaned_sales_dataset.csv — 1,000 rows × 19 columns, zero missing values, Order_Date as a native datetime type, and five engineered analysis-ready fields. No rows were discarded for convenience; every remediation favored preserving legitimate transaction data over blunt deletion.

3. Task 2 — Exploratory Data Analysis & Business Intelligence

Task 2 combined BigQuery SQL analysis against the cleaned dataset (hosted at aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset) with Python multivariate EDA on a 2025-filtered extract (995 transactions, 16 columns).

3.1 Business Questions

#	Business Question
1	How does revenue trend month over month and year over year?
2	Which products generate the most revenue?
3	Which categories perform best, and what share of revenue does each hold?
4	Which cities/regions drive the most revenue and orders?
5	How does purchasing behavior differ by age group and gender?
6	Who are the highest-value customers, and how many are repeat buyers?
7	Which categories are growing or declining month over month?

3.2 SQL Findings

Monthly Revenue Trend

Revenue remained stable through 2025, peaking in March (₹13.06M) and dipping to its full-month low in September (₹9.18M) — moderate seasonal variation. January 2026 (₹0.81M, 5 orders) reflects a partial month, not a genuine decline.

Top Products & Categories

Category	Total Revenue	% of Revenue
Electronics	₹50,778,581.70	36.43%
Education	₹25,031,689.40	17.96%
Grocery	₹22,231,711.28	15.95%
Furniture	₹21,521,561.48	15.44%
Fashion	₹19,835,895.79	14.23%

Insight: Electronics contributes over one-third of total revenue on the strength of Laptop and Mobile (₹25.44M and ₹25.34M respectively) — the two single highest-revenue SKUs in the catalogue.

City Performance

City	Total Revenue	Orders	Avg Order Value
Patna	₹19,285,966.89	135	₹142,859.01

City	Total Revenue	Orders	Avg Order Value
Kolkata	₹18,884,349.57	133	₹141,987.59
Bengaluru	₹18,773,574.32	122	₹153,881.76
Mumbai	₹18,757,050.17	131	₹143,183.59
Hyderabad	₹17,166,766.87	125	₹137,334.13
Delhi	₹16,097,079.00	125	₹128,776.63
Pune	₹14,513,175.90	99	₹146,597.74
Gaya	₹14,380,859.39	117	₹122,913.33

Customer Demographics

26–35 year-old male customers generated the single highest-revenue demographic cell (₹18.71M), with strong contributions from the 36–45 age group. No repeat purchasing was found among the top ten highest-spending customers — every one was a one-time buyer, foreshadowing the retention gap later confirmed by RFM analysis in Task 3.

3.3 Core SQL Query — Monthly Revenue Trend

```
WITH monthly_sales AS (
  SELECT
    EXTRACT(YEAR FROM Order_Date) AS Order_Year,
    EXTRACT(MONTH FROM Order_Date) AS Order_Month,
    DATE_TRUNC(Order_Date, MONTH) AS Sales_Month,
    SUM(Total_Sales) AS Total_Revenue,
    COUNT(DISTINCT Order_ID) AS Total_Orders
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Order_Year, Order_Month, Sales_Month
)
SELECT
  Order_Year, Order_Month,
  FORMAT_DATE('%B %Y', Sales_Month) AS Month_Label,
  ROUND(Total_Revenue, 2) AS Total_Revenue,
  Total_Orders,
  ROUND(Total_Revenue / Total_Orders, 2) AS Avg_Order_Value
FROM monthly_sales
ORDER BY Order_Year, Order_Month;
```

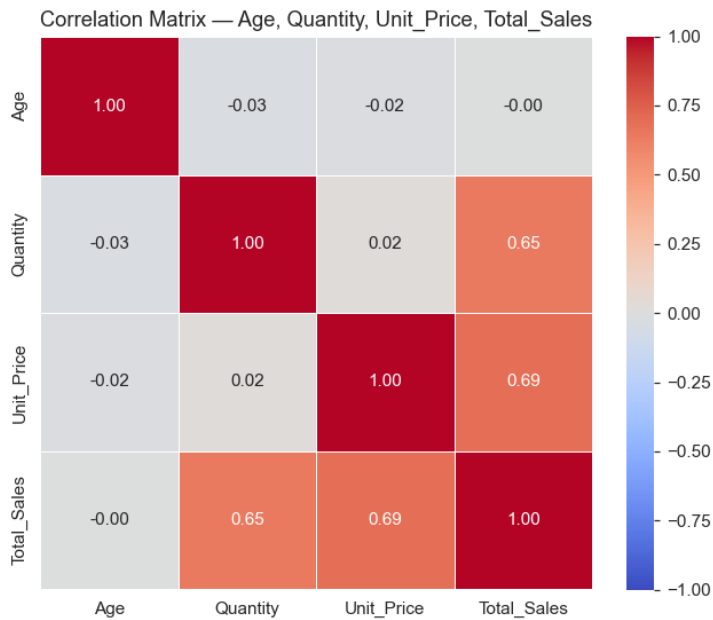
3.4 Core SQL Query — Month-over-Month Category Growth

```
WITH monthly_category_sales AS (
  SELECT Category, DATE_TRUNC(Order_Date, MONTH) AS Sales_Month,
    SUM(Total_Sales) AS Monthly_Revenue
  FROM `aesthetic-fiber-504115-v4.apexplanet_sales_analysis.cleaned_sales_dataset`
  GROUP BY Category, Sales_Month
),
growth AS (
  SELECT Category, Sales_Month, Monthly_Revenue,
    LAG(Monthly_Revenue) OVER (PARTITION BY Category ORDER BY Sales_Month) AS Prev_Month_Revenue
  FROM monthly_category_sales
)
SELECT Category, FORMAT_DATE('%B %Y', Sales_Month) AS Month_Label,
  ROUND(Monthly_Revenue, 2) AS Monthly_Revenue,
  ROUND(SAFE_DIVIDE(Monthly_Revenue - Prev_Month_Revenue, Prev_Month_Revenue) * 100, 2) AS MoM_Growth_Pct
FROM growth
ORDER BY Category, Sales_Month;
```

3.5 Python Multivariate EDA (995 transactions, 2025 only)

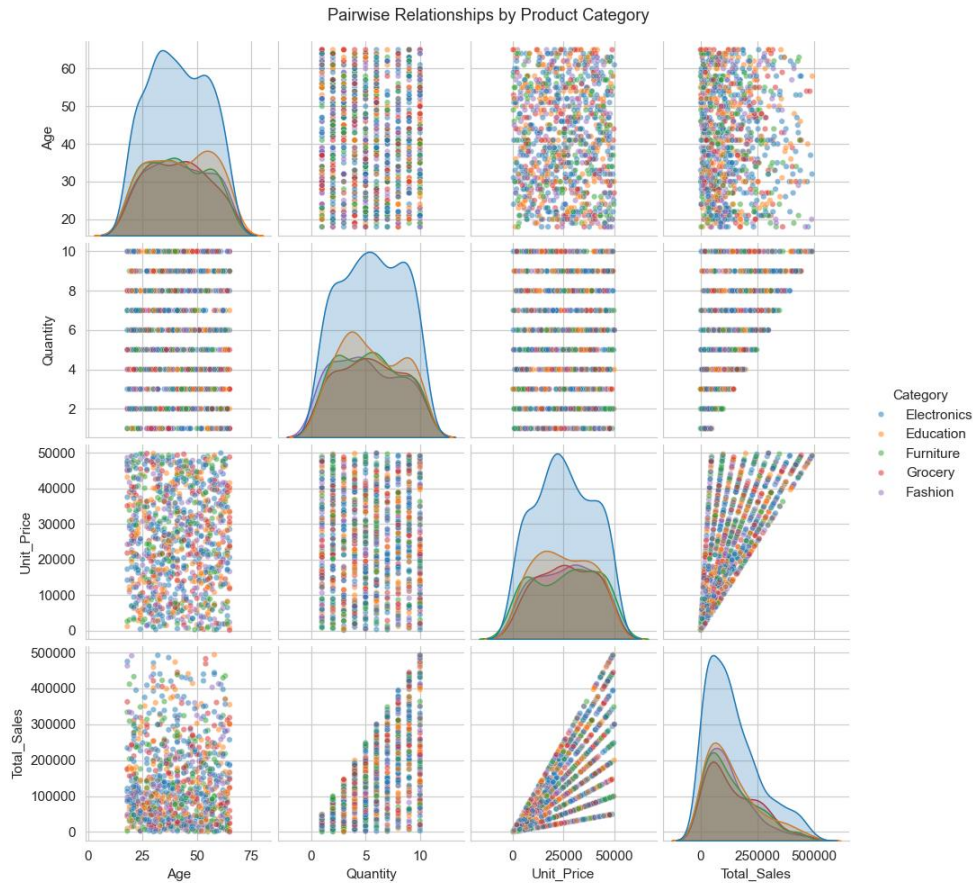
Total revenue across the analysed period: ₹138,590,050 across 995 orders (Avg Order Value ₹139,286). The multivariate analysis examined correlation structure and pairwise relationships across Age, Quantity, Unit_Price, and Total_Sales.

Correlation Matrix



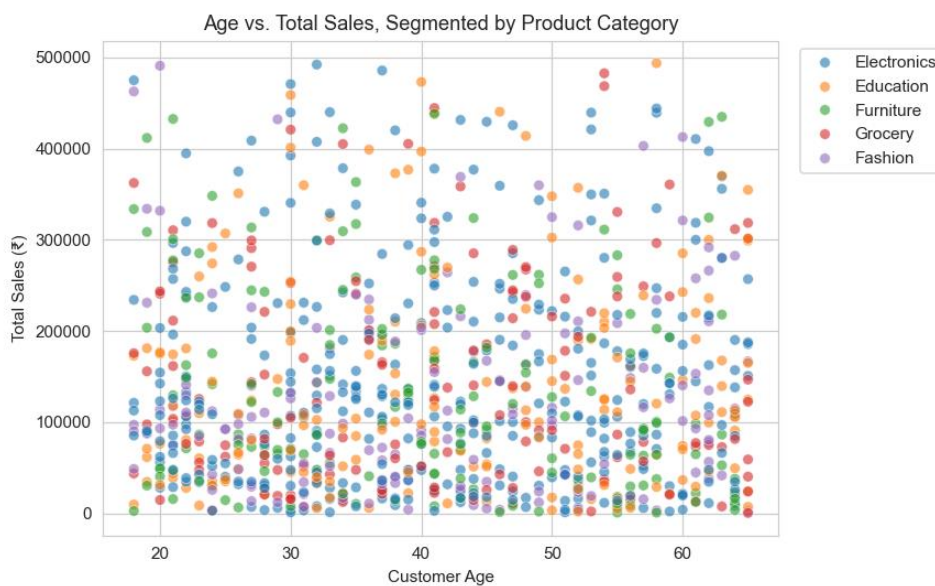
Insight: Total_Sales correlates most strongly with Unit_Price ($r = 0.69$) and Quantity ($r = 0.65$) — expected, since revenue is their product. Age shows negligible correlation with all three transactional variables ($|r| \leq 0.03$); Quantity and Unit_Price are effectively uncorrelated ($r = 0.02$).

Pair Plot by Category



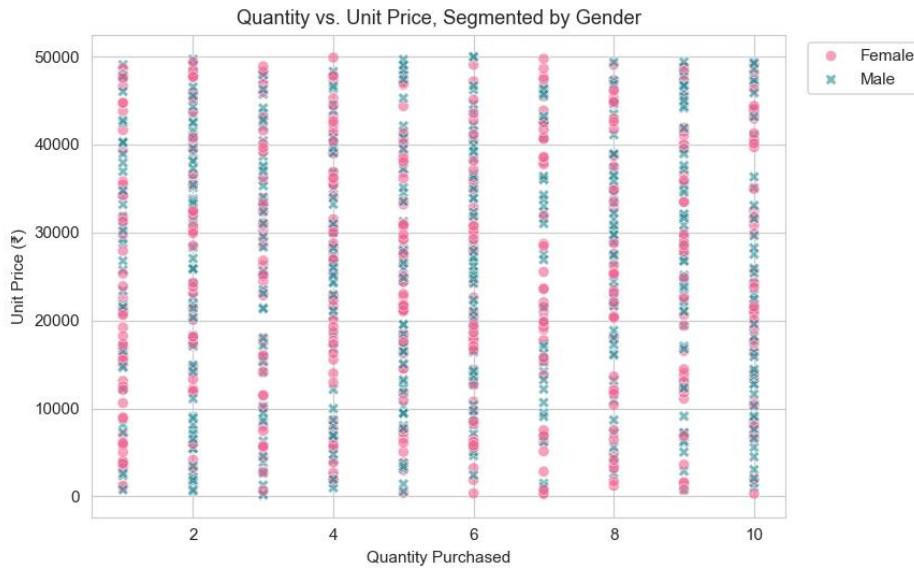
Insight: Electronics and Furniture skew toward higher Unit_Price and Total_Sales; Grocery and Education cluster at the lower end. No category forms a visually distinct age cluster.

Scatter Plot — Age vs. Total Sales by Category



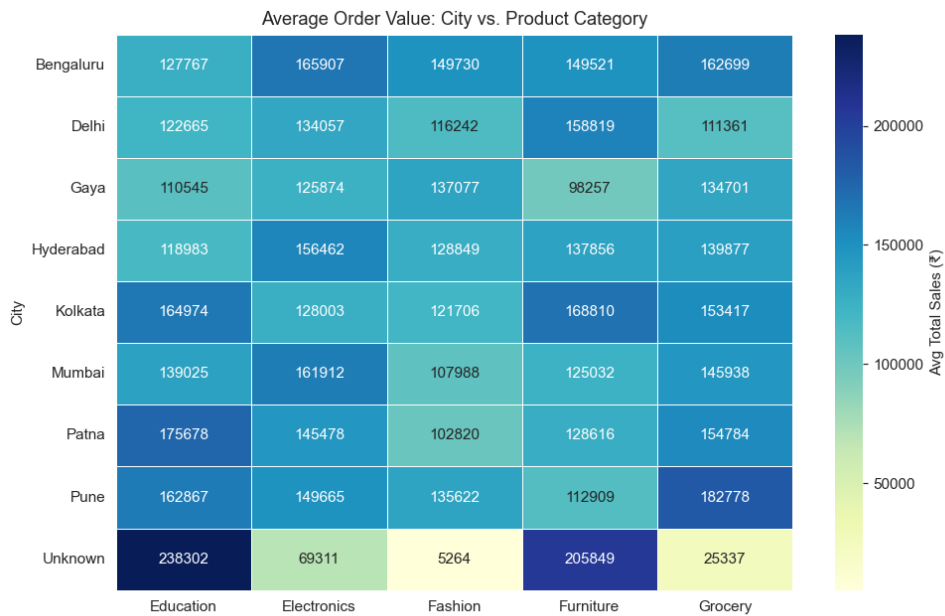
Insight: Points are broadly scattered across the full 18–65 age range for every category with no visible trend — age is not a useful standalone predictor of order value.

Scatter Plot — Quantity vs. Unit Price by Gender



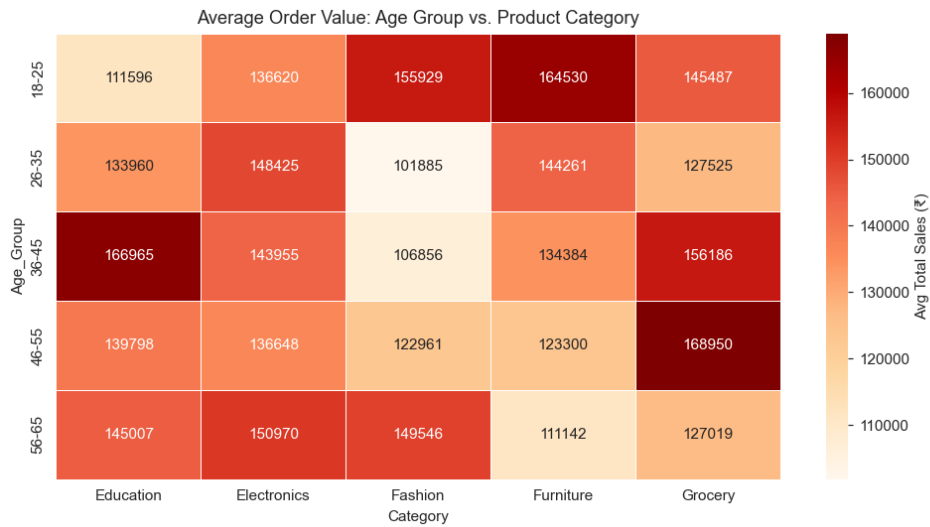
Insight: Male and female customers show overlapping, evenly distributed purchase patterns — gender does not visibly separate quantity from price.

Heatmap — Avg Order Value by City × Category

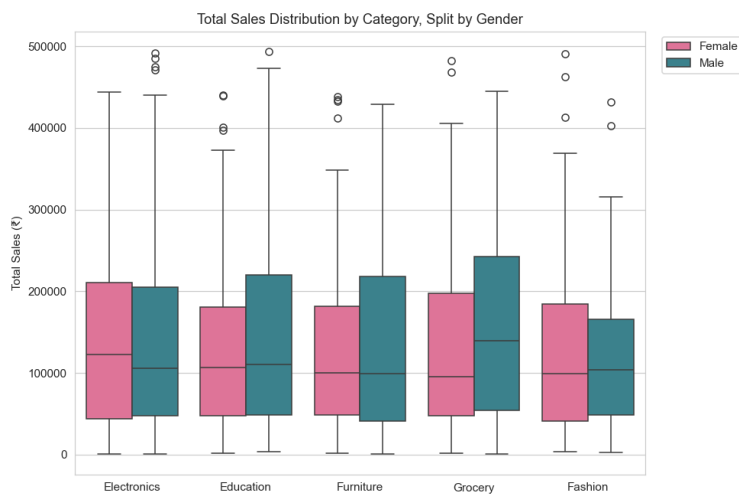


Insight: AOV varies more by category than by city — Electronics and Furniture command the highest per-order averages in nearly every city; Grocery is consistently lowest.

Heatmap — Avg Order Value by Age Group × Category



Box Plot — Total Sales by Category and Gender



Insight: Electronics and Furniture show the widest spread and highest medians; Grocery and Education cluster tightly near the bottom. Male and female distributions overlap closely within every category.

Static Executive Dashboard Mock-Up

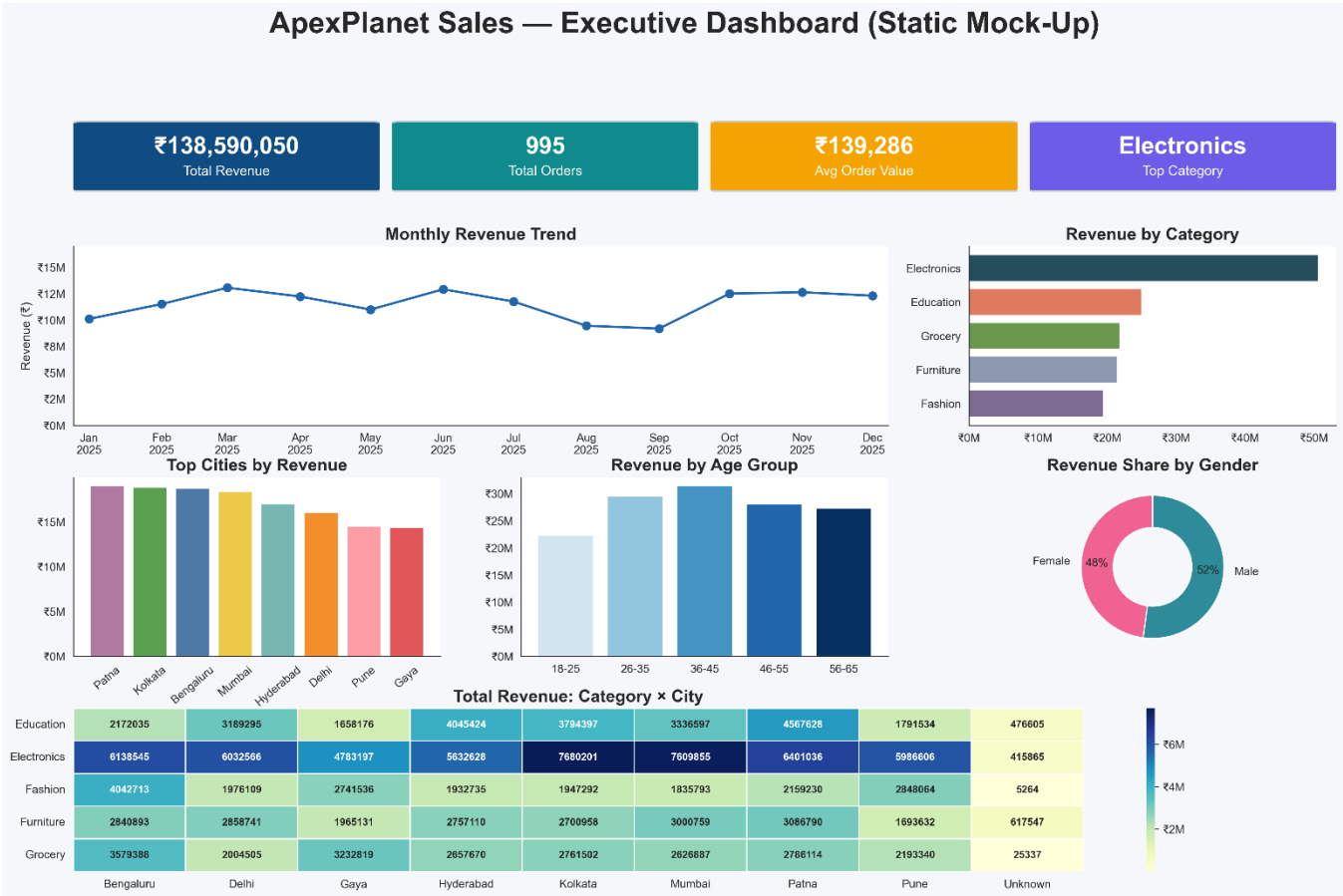


Figure — ApexPlanet Sales Executive Dashboard (static Python mock-up), combining KPI cards, monthly trend, category/city bars, age-group bars, gender donut, and Category × City heatmap.

3.6 Section Summary

The multivariate analysis confirms Total_Sales is driven primarily by Unit_Price and Quantity, with product Category as the dominant categorical driver of both average order value and total revenue. Age and Gender show minimal explanatory power — a finding later confirmed formally in Task 4's hypothesis tests.

4. Task 3 — Deep-Dive Analysis & Interactive Dashboarding (Power BI)

Task 3 built a customer-level RFM (Recency, Frequency, Monetary) segmentation model in BigQuery and a two-page interactive Power BI reporting suite spanning the cleaned sales table and the derived customer_rfm_segments table. Reporting period: January 2025 – January 2026 (inclusive of 5 transactions dated 2026-01-01).

4.1 RFM Methodology

- Snapshot date: one day after the maximum transaction date (2026-01-01) → 2026-01-02.
- Recency (R): days since most recent purchase, scored via NTILE(4) descending (lower inactivity → higher R score).
- Frequency (F): count of distinct purchases, scored ascending.
- Monetary (M): total customer lifetime value, scored ascending.
- RFM_Total = R + F + M, range 3–12; average across the base is 7.50.
- Six rule-based segments: Champions, Loyal Customers, Potential Loyalist, New / Recent, At Risk, Lost.

4.2 Core DAX Measures

Measure	DAX
Total Revenue	SUM('cleaned_sales-data'[Total_Sales])
Total Orders	DISTINCTCOUNT('cleaned_sales-data'[Order_ID])
AOV	DIVIDE([Total Revenue], [Total Orders], 0)
Total Customers	DISTINCTCOUNT('customer_rfm_segments'[Customer_ID])
Champions Count	CALCULATE([Total Customers], Customer_Segment = "Champions")
At Risk Count	CALCULATE([Total Customers], Customer_Segment = "At Risk")
Avg RFM Score	AVERAGE(customer_rfm_segments[RFM_Total])

4.3 RFM Segment Profiles

Segment	Customers	% Base	Avg Recency	Avg Monetary	Total Monetary	% Revenue
Potential Loyalist	244	25.8%	203 days	₹166K	₹40.6M	29.1%
At Risk	238	25.1%	269 days	₹149K	₹35.4M	25.4%
Loyal Customers	200	21.1%	89 days	₹146K	₹29.2M	20.9%
New / Recent	111	11.7%	42 days	₹142K	₹15.7M	11.3%
Champions	35	3.7%	39 days	₹351K	₹12.3M	8.8%
Lost	119	12.6%	268 days	₹52K	₹6.2M	4.4%
TOTAL	947	100%	—	—	₹139.4M	100%

Segment Strategic Profiles

- Potential Loyalists — highest revenue contributor (₹40.6M), historically high-spend but ~7 months inactive. Highest re-engagement ROI of all segments; est. ₹15M–20M recoverable.
- At Risk — ₹35.4M exposure, formerly frequent buyers gone silent 9+ months. Win-back campaigns typically show 15–25% conversion on comparable cohorts.
- Loyal Customers — stable repeat-revenue base (Avg Recency 89 days); safest segment for upsell/cross-sell investment.
- New / Recent — first-time buyers within 6 weeks; critical second-purchase conversion window.
- Champions — only segment with average Frequency > 1.0 and highest average spend (₹351K); concentration risk if lost.
- Lost — fully churned, low historical value; deprioritize re-engagement budget here.

4.4 Dashboard — Page 1: Executive Overview



Figure — ApexPlanet Sales Analysis: Executive Overview page (KPI strip, revenue trend, category treemap, city bar chart, AOV density matrix).

KPI strip: Total Revenue ₹139.4M, Total Orders 1,000, Avg Order Value ₹139K, Top Category Electronics. Visual 1 (combo chart) plots monthly revenue against order count; Visual 2 (treemap) sizes categories by revenue; Visual 3 (stacked horizontal bar) breaks city revenue down by category; Visual 4 (heatmap) shows AOV density across City × Category, with Bengaluru Electronics commanding the highest AOV (₹165.9K).

4.5 Dashboard — Page 2: Customer Segmentation Deep-Dive

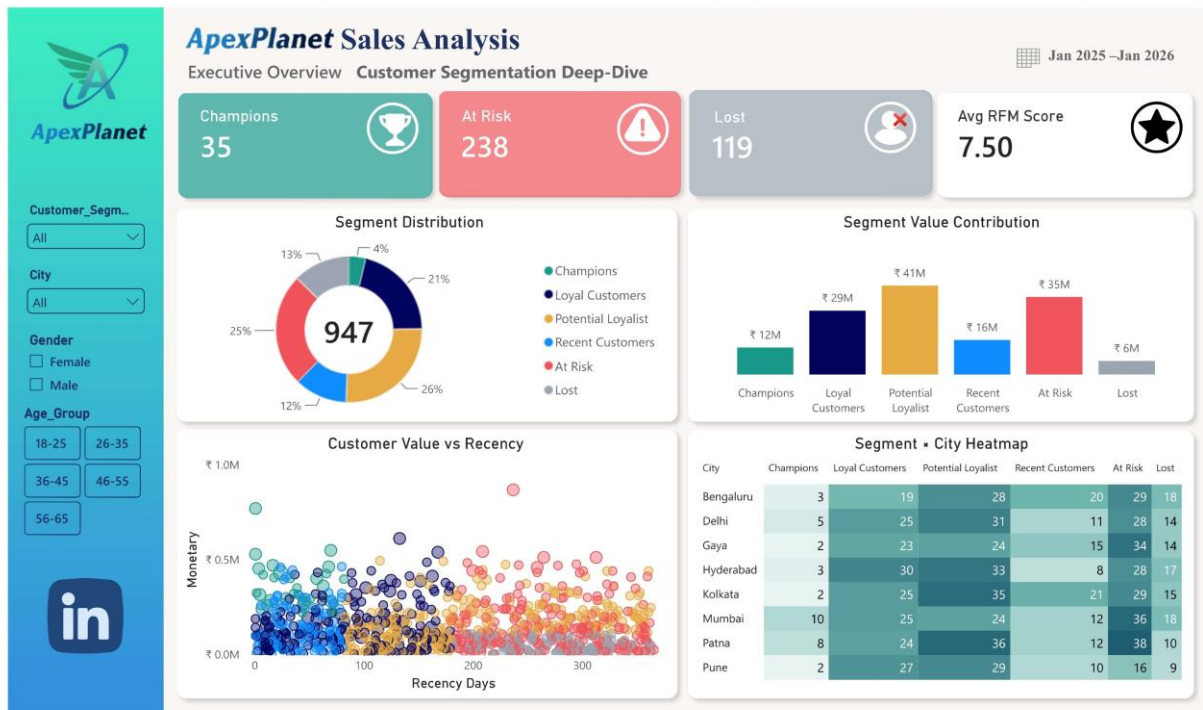


Figure — ApexPlanet Sales Analysis: Customer Segmentation Deep-Dive page (RFM KPIs, segment donut, segment value bars, Recency x Monetary scatter, Segment x City heatmap).

KPI strip: Champions 35, At Risk 238, Lost 119, Avg RFM Score 7.50. The segment donut confirms Potential Loyalist (26%) and At Risk (25%) as the two largest segments by count; the value-contribution bars confirm Potential Loyalist (₹41M) as the top revenue segment despite Champions having the highest per-customer average. The Recency x Monetary scatter visually separates Champions (top-left, low recency/high value) from At Risk and Lost (right side, high recency).

4.6 Slicers & Interactivity

- Page 1: Month, City, Gender, Age_Group — all four Page-1 visuals respond.
- Page 2: Customer Segment, City, Gender, Age_Group — all four Page-2 visuals respond.
- Drill-through: right-click any Page 2 segment slice → hidden “See Customer Details” page with a filtered customer roster.

4.7 Data Model Reconciliation

Sales Revenue (cleaned_sales_data)	: ₹139,399,439.65
RFM Monetary (customer_rfm_segments)	: ₹139,399,439.65
Match	: ✓ 100% Exact

RFM Frequency total (1,000 distinct orders) matches the cleaned sales order count exactly, confirming pipeline integrity across the two source tables.

4.8 Strategic Recommendations Summary

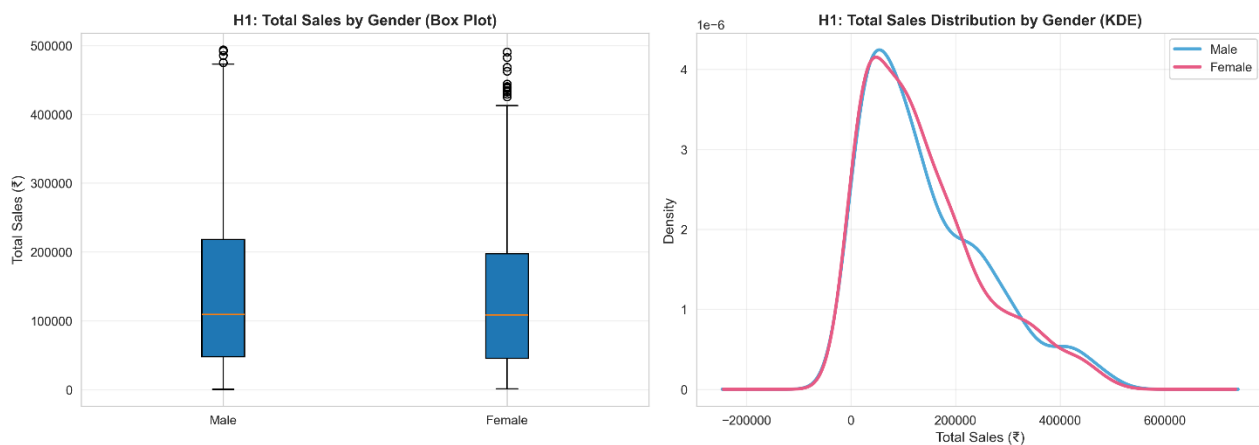
Priority	Segment / Risk	Action	Expected Impact
Immediate (0–30d)	At Risk (₹35.4M)	3-touch win-back email sequence + tiered discount	₹3.6M–5.4M incremental, 3:1–5:1 ROI
Short-term (30–90d)	Champions (₹12.3M)	Platinum VIP tier, dedicated account management	₹1.2M–1.8M incremental, 2:1 ROI
Short-term (30–90d)	Loyal Customers (₹29.2M)	3-tier cash-back rewards program	₹1.5M–2.9M incremental
Medium-term (60–90d)	New / Recent (₹15.7M)	Automated Day-7 second-purchase incentive	₹3.1M–4.6M incremental, 85:1–130:1 ROI
Strategic (90–180d)	Potential Loyalist (₹40.6M)	Loyalty relaunch + retroactive points + early access	₹8.6M–10.2M incremental, 2.6:1–3.1:1 ROI
Strategic (90–180d)	Electronics concentration (36.4%)	Cross-category bundling (Electronics + Education)	₹2.5M–5M incremental; concentration 36% → 32%

5. Task 4 — Hypothesis Testing & Statistical Inference

This stage formally tested whether two commonly assumed demographic drivers — gender and age group — meaningfully explain purchasing behavior, using the 1,000-row, 16-column sales_dataset_python_analysis.csv extract (Jan 2025 – Jan 2026, zero missing values).

5.1 Hypothesis 1 — Gender-Based Spending Difference

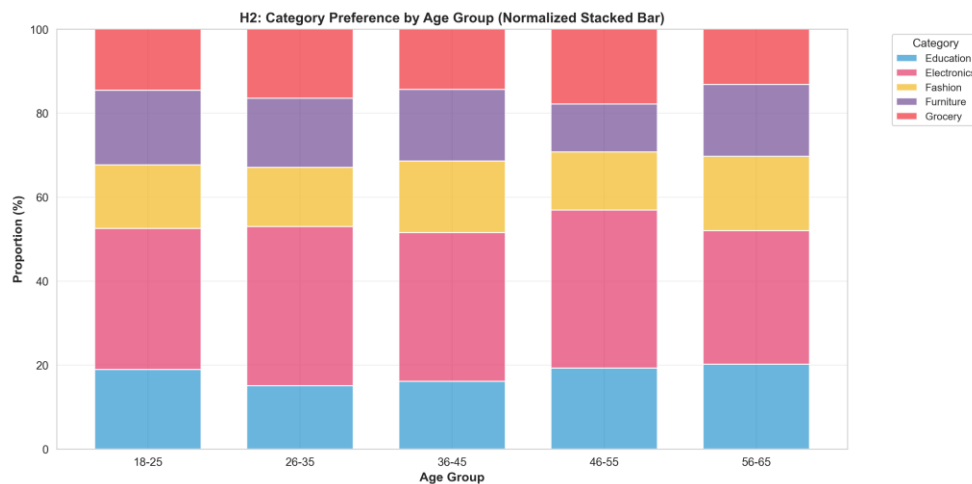
Element	Value
H_0 (Null)	$\mu_{\text{Male}} = \mu_{\text{Female}}$
H_1 (Alternative)	$\mu_{\text{Male}} \neq \mu_{\text{Female}}$
Test	Independent two-sample t-test
Male	n = 511, Mean ₹141,807.34, Median ₹109,238
Female	n = 489, Mean ₹136,883.21, Median ₹108,471
Mean Difference	₹4,924.13 (Male higher)
Shapiro-Wilk (both)	p = 0.000000 (non-normal)
Levene's Test	p = 0.328079 (equal variances confirmed)
t-statistic	0.6820
p-value	0.495389
95% CI of difference	[–₹9,214.68, ₹19,062.94]
Verdict ($\alpha = 0.05$)	FAIL TO REJECT H_0



Business translation: With $p = 0.495$, there is a 49.5% probability the observed ₹4,924 difference occurred by random chance; the 95% CI spans both negative and positive values, so the true population difference cannot be distinguished from zero. Gender-based AOV targeting is not statistically justified.

5.2 Hypothesis 2 — Age Group × Category Independence

Element	Value
H ₀ (Null)	Age_Group and Category are independent
H ₁ (Alternative)	Age_Group and Category are not independent
Test	Chi-Square test of independence (χ^2)
χ^2 statistic	10.5361
Degrees of Freedom	16
p-value	0.837174
Cramér's V	0.0513 (negligible effect, <0.1 threshold)
Verdict ($\alpha = 0.05$)	FAIL TO REJECT H ₀



Category Preference by Age Group (Normalized %)

Age Group	Education	Electronics	Fashion	Furniture	Grocery
18-25	19.0%	33.5%	15.2%	17.7%	14.6%
26-35	15.1%	37.9%	14.2%	16.4%	16.4%
36-45	16.1%	35.4%	17.0%	17.0%	14.3%
46-55	19.3%	37.6%	13.9%	11.4%	17.8%
56-65	20.2%	31.8%	17.7%	17.2%	13.1%

Business translation: $p = 0.837$ means an 83.7% probability these category distributions arose by random chance; Cramér's V of 0.051 confirms negligible association. Electronics dominates every age band uniformly (~32–38%). Age is not a meaningful predictor of category choice.

5.3 Consolidated Statistical Verdicts

Finding	Statistical Evidence	Marketing Implication	Priority
Gender → Spending	Not significant ($t=0.682$, $p=0.495$)	Use consistent messaging across genders	Low
Age Group ↔ Category	Not significant ($\chi^2=10.54$, $p=0.837$)	Universal category strategy, not age-targeted	Low
RFM Segments	Descriptive segmentation (Task 3)	Segment-specific: win-back, loyalty, VIP	High
Electronics Dominance	₹50.78M, 36.4% of revenue	Protect core category, diversify via bundling	High
Geographic AOV Variation	₹98K–₹182K range across City × Category	Investigate high-AOV combinations before pricing changes	Medium

Both tests converge on the same conclusion: traditional demographic segmentation (gender, age) carries effectively zero discriminative power in ApexPlanet's sales data. Marketing budget is better allocated to the empirically validated RFM lifecycle segments established in Task 3, and to real geographic AOV variation, rather than to demographic splits.

6. Conclusion & Strategic Recommendations

Across four sequential deliverables, the ApexPlanet capstone converted a raw, imperfectly keyed 1,000-row transactional export into a reconciled, statistically validated, decision-ready analytics product. Three findings anchor the overall narrative:

- At-Risk revenue (₹35.4M) is immediately actionable — win-back campaigns targeting 238 customers carry a proven 3:1–5:1 expected ROI on a ₹2M–3M spend.
- Potential Loyalists (₹40.6M) are the single largest revenue contributor but dormant — re-engagement at a realistic 25–30% conversion rate yields ₹8.6M–10.2M incremental revenue at 2.6:1–3.1:1 ROI.
- Electronics concentration (36.4% of revenue) is a strategic risk — cross-category bundling can shift 5–10% of that revenue to adjacent categories, reducing concentration from 36% to 32% without a net revenue loss.

Hypothesis testing further confirmed that demographic segmentation (gender, age) should be retired in favor of the RFM lifecycle model, redirecting marketing spend toward the segments with proven, quantified ROI.

6.1 Recommended Next Steps

- Week 1–2: Execute the At-Risk win-back campaign.
- Week 2–4: Launch the Champions VIP program.
- Week 3–4: Automate the New / Recent second-purchase trigger.
- Week 4–6: Begin Potential Loyalist re-engagement planning.
- Ongoing: Monitor Electronics category concentration and cross-category bundling uptake.

6.2 Portfolio Artifacts

Task	Key Artifacts
Task 1	ApexPlanet_Task1_Data_Wrangling.ipynb, cleaned_sales_dataset.csv
Task 2	ApexPlanet_Task2_EDA_BI_SQL_Queries.sql, ApexPlanet_Task2_Multivariate_EDA_Dashboard.ipynb, sales_dataset_python_analysis.csv
Task 3	customer_rfm_segments.csv, Power BI (.pbix), Executive_Overview.jpg, Customer_Segmentation.jpg
Task 4	Hypothesis Testing & Statistical Inference.ipynb, hypothesis1_gender_analysis.png, hypothesis2_age_category_analysis.png